



ELECT MAINT Department
SIMHADRI

Note No :1



Ref:SMTTP/ELECT MAINT/2024-25/LMIDC S/824003

Date:

04/01/2025(19:14:47)

Subject:Reg. approval of 3rd revision of LMIODOPSELEC002
(MANAGEMENT OF DC SUPPLY SYSTEM FAILURE)

3rd Revision of LMI/OD/OPS/ELEC/002 (MANAGEMENT OF DC SUPPLY SYSTEM FAILURE) is proposed through this note.

LMI page nos. are reviewed and corrected as per External LMI Audit observation.

Operation and C&I is requested to review the proposed revision of the LMI.

Put up for further review and approval.

Submitted by:

राजेन्द्र सिंह - उप महाप्रबंधक

RAJENDRA SINGH - 062078 (DY. GENERAL MANAGER ,ELECT MAINT) ,Simhadri
(RAJENDRASINGH@NTPC.CO.IN, 9650995459)

On:04/01/2025(19:14:47) **Ref:**SMTTP/ELECT MAINT/2024-25/LMIDC S/824003

Note No:2

Checked the docuemnt. May please approve.

Submitted by:

मेडा नाग राजा - उप महाप्रबंधक

Meda Naga Raja - 008404 (DY. GENERAL MANAGER ,ELECT MAINT)
(MEDANAGARAJA@NTPC.CO.IN, 9437578452) ,Simhadri

On:08/01/2025(11:54:10) **Ref:**SMTTPP/ELECT MAINT/2024-25/LMIDC S/824003

Note No:3

Forwarded for further approval

आगे की स्वीकृति के लिए भेजा गया

Submitted by:

तेन्नेटि विजय कृष्ण - अपर महाप्रबंधक

Tenneti Vijay Krishna - 007966 (ADDL. GENERAL MANAGER ,ELECT MAINT)
(TVIJAYKRISHNA@NTPC.CO.IN, 9437188892) ,Simhadri

On:10/01/2025(08:48:19) **Ref:**SMTTPP/ELECT MAINT/2024-25/LMIDC S/824003

Note No:4

Please review

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri

On:10/01/2025(10:17:56) **Ref:**SMTPP/ELECT MAINT/2024-25/LMIDC S/824003

Note No:5

Please add LMI Check list as Annexure as last part of LMI

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

On:11/01/2025(10:56:44) **Ref:**SMTPP/ELECT MAINT/2024-25/LMIDC S/824003

Note No:6

LMI check list is added as Annexure. Please process.

Submitted by:

राजेन्द्र सिंह - उप महाप्रबंधक

RAJENDRA SINGH - 062078 (DY. GENERAL MANAGER ,ELECT MAINT)
(RAJENDRASINGH@NTPC.CO.IN, 9650995459) ,Simhadri

On:11/01/2025(20:07:42) **Ref:**SMTTPP/ELECT MAINT/2024-25/LMIDC S/824003

Note No:7

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

On:13/01/2025(12:04:31) **Ref:**SMTTPP/ELECT MAINT/2024-25/LMIDC S/824003

Note No:8

Reviewd as discussed.

Submitted by:

राजेन्द्र सिंह - उप महाप्रबंधक

RAJENDRA SINGH - 062078 (DY. GENERAL MANAGER ,ELECT MAINT)
(RAJENDRASINGH@NTPC.CO.IN, 9650995459) ,Simhadri

On:13/01/2025(18:49:03) **Ref:**SMTTPP/ELECT MAINT/2024-25/LMIDC S/824003

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Reviewed

Submitted by:

अजय गुप्ता - उप महाप्रबंधक

Ajay Gupta - 008565 (DY. GENERAL MANAGER ,EEMG)
(AJAYGUPTA04@NTPC.CO.IN, 9958105993) ,Simhadri

On:15/01/2025(17:25:45) **Ref:**SMTTPP/ELECT MAINT/2024-25/LMIDC S/824003

Note No:10

LMI is in line with the OD.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri

On:17/01/2025(12:00:43) **Ref:**SMTTPP/ELECT MAINT/2024-25/LMIDC S/824003

Note No:11

May pl review the LMI

Submitted by:

जे आर वर्गीस - अपर महाप्रबंधक

Jayamol Rachel Varughese - 006127 (ADDL. GENERAL MANAGER ,C & I MAINT)
(JAYAMOLRV@NTPC.CO.IN, 9440918557) ,Simhadri

On:17/01/2025(15:53:30) **Ref:**SMTPP/ELECT MAINT/2024-25/LMIDC S/824003

Note No:12

Reviewed

Submitted by:

अजीतकुमार रौशन - वरिष्ठ प्रबंधक

AJEET KUMAR RAUSHAN - 103000 (SENIOR MANAGER ,C & I MAINT)
(AKRAUSHAN@NTPC.CO.IN, 9434077809) ,Simhadri

On:17/01/2025(17:33:17) **Ref:**SMTPP/ELECT MAINT/2024-25/LMIDC S/824003

Note No:13

Forwarded for approval after review

Submitted by:

जे आर वर्गीस - अपर महाप्रबंधक

Jayamol Rachel Varughese - 006127 (ADDL. GENERAL MANAGER ,C & I MAINT)
(JAYAMOLRV@NTPC.CO.IN, 9440918557) ,Simhadri

On:21/01/2025(09:42:18) **Ref:**SMTPP/ELECT MAINT/2024-25/LMIDC S/824003

Note No:14

Please review.

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

Somnath Bhattacharya - 007203 (ADDL. GENERAL MANAGER ,OPERATION)
(SOMNATHBHATTACHARYA@NTPC.CO.IN, 9431600725) ,Simhadri

On:24/01/2025(17:40:54) **Ref:**SMTPP/ELECT MAINT/2024-25/LMIDC S/824003

Note No:15

Proposed LMI is reviewed and found in line with OD.

Submitted by:

मलय कांति जाना - अपर महाप्रबंधक

Malay Kanti Jana - 007824 (ADDL. GENERAL MANAGER ,OPERATION)
(MKJANA@NTPC.CO.IN, 9437963403) ,Simhadri

On:27/01/2025(10:50:45) **Ref:**SMTPP/ELECT MAINT/2024-25/LMIDC S/824003

Note No:16

Checked and forwarded.

Submitted by:

सोमनाथ भट्टाचार्य - अपर महाप्रबंधक

Somnath Bhattacharya - 007203 (ADDL. GENERAL MANAGER ,OPERATION)
(SOMNATHBHATTACHARYA@NTPC.CO.IN, 9431600725) ,Simhadri

On:31/01/2025(16:21:41) **Ref:**SMTPP/ELECT MAINT/2024-25/LMIDC S/824003

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Reviewed and submitted for approval

Submitted by:

ओम प्रकाश - अपर महाप्रबंधक

Om Prakash - 007105 (ADDL. GENERAL MANAGER ,MAINTENANCE)
(OMPRAKASH08@NTPC.CO.IN, 9650994070) ,Simhadri

On:03/02/2025(17:34:56) **Ref:**SMTTPP/ELECT MAINT/2024-25/LMIDC S/824003

Note No:18

LMI is in line with the OD.

May please be approved.

Submitted by:

नवीन किशोर - अपर महाप्रबंधक

Navin Kishore - 006964 (ADDL. GENERAL MANAGER ,EEMG)
(NAVINKISHORE@NTPC.CO.IN, 9650990460) ,Simhadri

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May Please be approved.

Submitted by:

प्रसन्ता कुमार जेना - महाप्रबंधक

Prasanta Kumar Jena - 006081 (GENERAL MANAGER ,O & M)
(PKJENA@NTPC.CO.IN, 9650990836) ,Simhadri

On:09/02/2025(13:15:26) **Ref:**SMTTPP/ELECT MAINT/2024-25/LMIDC S/824003

Note No:20

Approved & Submitted by:

समीर शर्मा - मुख्य महाप्रबंधक

Sameer Sharma - 005041 (CGM ,HEAD OF PROJECT)
(SAMEERSHARMA@NTPC.CO.IN, 9650990022) ,Simhadri

On:10/02/2025(19:47:43) **Ref:**SMTTPP/ELECT MAINT/2024-25/LMIDC S/824003

NTPC

SIMHADRI

LOCATION MANAGEMENT INSTRUCTION

MANAGEMENT OF DC SUPPLY SYSTEM FAILURE

LMI/OD/OPS/ELEC/002 Rev:3

JAN - 2025

APPROVED FOR IMPLEMENTATION BY HOP (SIMHADRI)

LMI APPROVAL DOCUMENT

1	LMI Name	MANAGEMENT OF DC SUPPLY SYSTEM FAILURE
2	LMI No	LMI/ OD/OPS/ELEC/002, REV:3
3	Reference Document	COS-OD/ OPS/ELEC/002, REV:1
4	Superseded Document	LMI/ OD/OPS/ELEC/002, REV:2
5	Prepared by	Rajendra Singh DGM (EM)
6	Checked by	Meda Naga Raja DGM (EM)
7	Checked by	Tenneti Vijay Krishna AGM(EM)
8	Reviewed by	Jayamol Rachel Varughese AGM (C&I)
9	Reviewed by	Somnath Battacharya AGM (Operation)
10	Reviewed by	Om Prakash AGM(Maint)
11	Reviewed by	Prasanta Kumar Jena GM(O&M)
12	Approved by	Sameer Sharma HOP (Simhadri)

TITLE	MANAGEMENT OF DC SUPPLY SYSTEM FAILURE
LMI NO	LMI/OD/OPS/ELEC/002, Rev:3

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TITLE	MANAGEMENT OF DC SUPPLY SYSTEM FAILURE
LMI NO	LMI/OD/OPS/ELEC/002, Rev:3

1. INTRODUCTION:

DC supply system within a Power Station is of paramount importance and its failure could lead to catastrophic consequences due to protection system and vital control system being rendered inoperative.

Even though the failure of DC system is unlikely, in view of the consequences of such a failure, it is very much essential to address the systems review and gear up for action plan as regards the shutdown procedures adopted and/or the corrective action taken in case of occurrence of such an incident.

2. SCOPE:

This LMI is applicable to

- i) 220 V DC System of Unit # 1,2,3,4
- ii) 24 V DC System of Unit # 1,2
- iii) 220 V DC System of 400KV Switch yard
- iv) 48 V DC System of 400KV Switch yard
- v) 220/110 V/24V DC System of MUW area
- vi) UPS system of Unit # 1,2,3,4
- vii) 24V DC System of Stg-2 SG (Unit-3,4)
- viii) 24V DC System of Stg-2 TG (Unit-3,4)
- ix) 24V DC System of Stg-2 Station C&I (Unit-3,4)
- x) 24V DC System of offsite area inside the plant. Stg-2 CWPH
- xi) 24V DC System of Stg-2 CPU Regeneration plant.
- xii) 24V DC System of Stg-1& 2 DM plant
- xiii) 24V DC System of Stg-2 FOPH
- xiv) 24V DC System of U# 3,4 ESP
- xv) 24V DC System of Stg-2 FWPH
- xvi)** 24V DC System of Stg-2 Ash Handling plant
- xvii) 24V DC Sweet pump house
- xviii) 24V DC Raw pump house
- xix) 24V DC AWRC pump house
- xx) 24V DC Sea water pump house

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3. REFERENCE DOCUMENTS:

Operation directive: OD/OPS/ELEC/002 Rev No.:1

4. SUSPENDED DOCUMENTS:

LMI/ OD/OPS/ELEC/002 Issue: 01 Rev: 2 Dated: Aug 2023

5. DETAILS OF REVISION:

LMI page nos. are reviewed and corrected as per External LMI Audit observation.

6. Possibilities of DC failures & reviving procedures:

The following are the main causes of D.C failures:

1. Faults in battery and battery charger.
2. Overheating of D.C.D.B modules.

List of different DC banks and chargers in NTPC Simhadri and their failure and reviving procedure tabulated below.

TITLE	MANAGEMENT OF DC SUPPLY SYSTEM FAILURE
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Unit 220 V DC Failure

Stage-1&2 (4*500 MW Units)

DC Supply Configuration	220 Volts DCDB is in two sections, each section is provided with one battery charger and one set of battery bank. Normally the bus coupler remains in open condition. During normal operation, battery charger feeds the DC load and battery will be in floating condition. Whenever the charger is to be taken for maintenance or the battery is to be put in boost mode, both charger and battery will be disconnected from load side after ensuring that the both sections load is fed through other charger and battery bank. 1. When one battery bank or charger is to be isolated, it should be done in presence of Electrical Maintenance (EM) with the consent of Shift Charge Engineer (SCE). 2. It may be ensured that there is no DC Earth fault in any of the sections before interconnecting them through bus coupler.
Possible situations for Non-availability of DC Supply	Failure of total DC supply of Unit 1. Flash over in DCDB 2. DCDB both the incomer breakers getting tripped 3. Fire in Battery Bank 4. Battery chargers trouble may be due to non-availability of AC supply or any faults in chargers.
Annunciations / Alerts	▪ DCDB Sec-1 &2 under voltage Alarm appears in MMI. ▪ All HT Breaker Status indications will become bad ▪ All LT breaker status indication will become bad ▪ Battery charger trouble alarm appear in MMI.
Consequences of DC Supply fail (Situation wise)	1. Boiler will trip on MFT after 2 sec due to 220 V DC failures. All HT drives like fans, mills, CC pumps will continue to run as their breakers cannot trip without DC. 2. Turbine will trip on MFT protection. 3. Generator trip command will be initiated from turbine trip but since 220V DC is not available, GCB cannot be tripped electrically (for Unit-1 &2 only). 4. DC drives cannot come into service. 5. Generator, GCB (for Unit-1&2 only), GT protection tripping as well as electrical operation will not be available. Generator will be running as synchronous motor. HP/IP/ LP casing exhaust temperature will start increasing

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	6. 400 KV breakers can be operated.
Steps for Safe Isolation / Shutdown of plant in case re-establishment of DC source is not feasible.	1. Make open announcement in the PA system and alert all concerned. Resp-Unit Control Engineer(UCE) 2. Manually trip GCB (for Unit-1 & 2 only) and field breaker from the local panel. Resp-UCE 3. Stop the necessary HT drives through switchgear manual push button. Resp- UCE 4. Identify the cause of DC failure and restore the same with the help of Elect Maint. Resp- UCE Ensure the following before opening 400KV main and tie breakers from 400KV Switchyard. 1. Start DG Set. Resp-TG Desk Engineer 2. Charge emergency switchgear through DG set by tripping the normal incomer breaker from IDA/2DA manually and closing the DG incomer manually from local breaker push buttons. Resp-TG Desk Engineer 3. Start Main Turbine AOP-I which is in Unit emergency bus. Resp-TG Desk Engineer 4. Start AC Seal Oil Pump-1 which is in Unit emergency bus. Resp-TG Desk Engineer 5. Start AOP- A of all TDBFPs which is in Unit emergency bus. Resp-TG Desk Engineer 6. Inform switchyard engineer to trip 400 KV side main and tie breakers. Resp- UCE and Switchyard Engr. This will result in total power failure in the unit with emergency switchgear in charged condition through DG set. Take actions as

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	per the total power supply failure procedure with special care on emergency switchgear equipments as D.C. lube oil and seal oil pumps will not be available. Resp: UCE
Methodology for re-establishing the DC Supply	Inform EM and associate for restoration of supply to DCDB at the earliest. Resp: SCE / AGM(EM)
Modifications Recommended	NIL

Failure of Un-interrupted Power Supply (UPS) in Stage 1&2

DC Supply Configuration	1. In this system there are two UPS chargers with one common battery bank of 400 volt. 2. The chargers are feeding supply to two inverters. Normally both the inverters are in service and if one inverter fails then other inverter takes the load. 3. If both the invertors fail, then one static bypass emergency supply comes in service. If the above static bypass switch also does not come into service then the total UPS failure occurs.
Possible situations for Non-availability of UPS Supply	Failure of DC to both Inverters and alternate source fail to come in service.
Consequences of UPS fail	• Boiler will trip • Boiler will trip on flame failure as flame scanner supply is from UPS. • All Gravimetric Feeders will trip on loss of UPS as Feeder Micro Processor Card Power Supply is derived from UPS. • This will in turn result in loss of all fuel Trip of Boiler in case Oil Guns are not in Service at that moment of time. • On all BFPs trip (due to UPS failure D/A Level very low Switches will act) for stage-1

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	• MDBFP scoop from UPS supply which remains in stay put for stage 1&2 • Turbine will trip due to MFT and generator will trip due to Turbine tripping / low forward Power protection. • MMI Servers and Work Stations will go into hard shut down mode. • No commands can be executed from MMI and process parameters will not be available. • Vertical panel will be available for operation in stage-1& UCD will be available for stage-2 • But some of process indications will not be available in the vertical panel also.
Steps for safe Isolation / shutdown of plant in case re-establishment of UPS source is not feasible.	1. Start main Turbine AOP/JOP / AC Seal Pumps from vertical control panel/UCD as direct start provision is available for them. Resp-TG Desk Engineer 2. Check the status of all critical equipments and ensure safe shutdown whenever required. Resp- UCE
Methodology for re-establishing the UPS Supply	Restore UPS at the earliest in association with C&I/EM Resp: SCE / DGM (EM) / DGM (C&I)
Recommendations / Modifications	Nil

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Unit 24 V DC Failure**Stage-I (2*500 MW Units)**

DC Supply Configuration	24 Volts DCDB is in two sections, each section is provided with one battery charger and one set of battery bank. Normally the bus coupler remains in open condition. During normal operation, battery charger feeds the D.C load and battery will be in floating condition. Whenever the charger is to be taken for maintenance or the battery is to be put in boost mode, they will be disconnected from load side after ensuring that the both sections load is fed through other charger and battery bank. All C&I control panels have 24V supply auto change over provision through diode coupling option. In case of one supply interruption, other one will take over, and an alarm will be initiated. When one Battery bank or charger is to be isolated, it should be done in presence of C&I and EM with the consent of SCE.
Possible situations for Non-availability of DC Supply	Failure of 24V DC supply Flashover in 24V distribution MCC
Annunciations / Alerts	All indications will become blue in MMI and Vertical panel indications will be lost.
Consequences of DC Supply fail (Situation wise)	1. No C&I command can be executed. No equipment remote operation is available. 2. Boiler will trip on total 24 V DC failure trip through hard wiring. 3. All HT drives like PA fans & mills will continue to run as their breakers trip command will not extend to switchgear without 24 volts DC. Gravimetric feeders will trip on MFT. 4. ID fans both VFD channels will trip. 5. Turbine will not trip without 24 volts DC supply. 6. Generator will be in connected condition to grid.

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	7. All Drives with speed controls like TDBFP will come to minimum. 8. PA fans, FD fans blade pitches will be at last position(stay put) 9. CEP and BFP recirculation valves will open. 10. EHC command will be at maximum, hence turbine control valves will be full open and load will shoot up to maximum possible with valve wide open condition. Lube oil temperature, primary water, cold gas temperature, HP control fluid temperature control valves remain stay put. 11. HP heaters/LP heaters normal drip control valves will close and alternate drip control valves will open. 12. Vacuum breaker will not operate. 13. Turbine exhaust spray valve will not operate.
Steps for Safe Isolation / Shutdown of plant in case re-establishment of DC source is not feasible.	1. Main Turbine AOP-I, ACSOP-I & JOP-1 should be started from vertical panel as direct start provision is available for them. Resp-TG Desk Engineer 2. Monitor seal oil and lube oil systems parameters. Resp-TG Desk Engineer 3. Turbine is to be tripped from local trip gear in the governing rack. Resp-UCE 4. After tripping of turbine, Generator is to be tripped from GRP panel by activating GRP relay (low forward power relay 37 G1 or G2). Resp-TG Desk Engineer 5. PA fans are to be tripped on top most priority from local switchgear as remote tripping is not possible. Resp- Electrical operator 6. After ensuring the tripping of PA Fans, trip FD fans from local switchgear as remote tripping is not possible. Resp- Electrical operator 7. After tripping PA and FD fans, mills, CC Pumps, CEP, etc are to be tripped from local switchgear as remote tripping is not possible. Resp- Electrical operator 8. TDBFPs are to be tripped from local to avoid drum level high problem. Resp-TG Floor operator

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	9. TDBFP & Main Turbine barring gear valve to be opened from local. Resp-TG Floor operator
Methodology for re-establishing the DC Supply	Take help from CI & EM for supply restoration. Resp: SCE / AGM (EM) / AGM (C&I)
Recommendations / Modifications	

TG -24 V DC Failure
Stage-II (2*500 MW Units)

DC Supply Configuration	Each Unit is having two independent 24 V DC boards for TG with their own Battery Banks (670AH) & two Battery chargers for each DCDB section. Charger-1A, 2A from UES and Charger-1B,2B from USS Sec-B. Bus coupler is available between both the boards.
Possible situations for Non-availability of DC Supply	Failure of both 24V DC supply Flashover in TG 24V DCDB
Annunciations / Alerts	Panel 24 v power supply failures are annunciated in the control room on the LVS

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Consequences of DC Supply fail (Situation wise)	1. No C&I command can be executed. No equipment remote operation is available. 2. Turbine will not trip without 24 volts DC supply. 3. Generator will be in connected condition to grid. 4. All Drives with speed controls like TDBFP will come to minimum. 5. CEP and BFP recirculation valves will open. 6. EHC command will be at maximum, hence turbine control valves will be full open and load will shoot up to maximum possible with valve wide open condition. Lube oil temperature, primary water, cold gas temperature, HP control fluid temperature control valves remain stay put. 7. HP heaters/LP heaters normal drip control valves will close and alternate drip control valves will open. 8. Vacuum breaker will not operate. 9. Turbine exhaust spray valve will not operate. 10. BFP Exhaust spray will not operate. 11. HP/LP Bypass will not operate. 12. TG to Gen Trip will not operate. 13. MFT to Turbine will not operate.
Steps for Safe Isolation / Shutdown of plant in case re-establishment of DC source is not feasible.	1. Start DC EOP from direct start push button given on control desk 2. Trip boiler by pressing Emergency trip buttons on Control Desk. All Mills, Feeders & both PA fans will trip. 3. Trip Turbine from local trip device in TG Governing Rack 4. Generator will trip on Low forward power protection after 30 seconds. Otherwise open both 400KV Breakers from ECD. 5. Stop BCW Pumps from MMI if drum level is too low. 6. TDBFPs are to be tripped from MMI to avoid drum level going high 7. Close MS Stop Valves MS-1&2 from MMI 8. Monitor condenser vacuum as both vacuum pumps inlet valves will close 9. Start DC JOP from local panel (Starter Cabinet) 10. Start AOP from Switch gear module 11. Barring gear valves of main turbine & TDBFPs are to be opened from local
Methodology for re-establishing the DC Supply	Inform C&I & EM for early restoration of DC Supply. Resp: SCE / AGM (EM) / AGM (C&I)
Recommendations / Modifications	NIL

SG 24V DC Failure

TITLE	MANAGEMENT OF DC SUPPLY SYSTEM FAILURE
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Stage-II (2*500 MW Units)

DC Supply Configuration	Each Unit is having two independent 24 V DC boards for SG with their own Battery Banks (1030AH) & two Battery chargers for each DCDB section ; Charger-1A,2A from UES and Charger-1B,2B from USS Sec-B. Bus coupler is available between both the boards.
Possible situations for Non-availability of DC Supply	Failure of total DC supply of Unit 1. Flash over in DCDB (Total loss of DC supply will not happen in any other situations)
Annunciations / Alerts	DCDB Sec-A & DCDB Sec-B voltage Abnormal alarm appears, DCDB volts read zero
Consequences of DC Supply fail (Situation wise)	No C&I command can be executed. No equipment remote operation is available. 1. Boiler will trip on total 24 V DC failure trip through hard wiring. 2. (a) HT drive FD fans will continue to run as their breakers trip command will not extend to switchgear without 24 volts DC. (b) Mills, Gravimetric feeders will trip. 3. ID fans both VFD channels & PA fans will trip. 4. FD fans blade pitches will be at minimum position
Steps for Safe Isolation / Shutdown of plant in case re-establishment of DC source is not feasible.	1. All mills will trip on hard wired logic from UCD 2. Trip both PA Fans from MMI 3. Trip both FD fans from MMI to avoid furnace pressurization as both ID fans speed will be driven to minimum on 24V supply failure 4. Trip Turbine from Emergency Trip push button on control desk 5. Generator will trip on Low forward power protection 6. Monitor TG Lube Oil & Seal oil system 7. parameters
Methodology for re-establishing the DC Supply	Inform EM and C&I for restoration of supply to DCDB at the earliest. Resp: SCE / DGM (EM) / DGM (C&I)

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Modifications Recommended	NIL
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STATION C&I 24 V DC Failure Stage-II (2*500 MW Unit)

DC Supply Configuration	Each Unit is having two independent 24V DC boards for Station C&I with their own Battery Banks (1600AH) & two Battery chargers for each DCDB section ; Charger-1A,2A from UES and Charger-1B,2B from USS Sec-B. Bus coupler is available between both the boards.
Possible situations for Non-availability of DC Supply	Fire in Chargers room which contains station C&I chargers
Annunciations / Alerts	Panel 24 v power supply failures are annunciated in the control room on the LVS
Consequences of DC Supply fail (Situation wise)	All HT, LT breaker Indication and operation will not be available from HMI/Back panel Rack. Blade pitch of PA & FD fan will be at stay put.
Steps for Safe Isolation / Shutdown of plant in case re-establishment of DC source is not feasible.	1. ID fans VFDs speed will be driven to minimum and BFPs recirculation valves will open 2. Boiler will trip either on drum level low or furnace pressure high 3. Turbine will trip on MFT to turbine trip. 4. Generator will trip on low forward power protection 5. Both 11KV UA/UB buses will become dead resulting in loss of all unit auxiliaries. All Switchgear breakers will become inoperative from MMI. 6. Start DC EOP & DC SOP from direct start push buttons given on control desk. 7. Trip both TDBFPs from Local trip device. 8. Ensure ID/FD/PA fans & BCW Pumps are in tripped condition from switchgear. Otherwise trip them from switchgear emergency trip levers. 9. Start DG Set from Local and charge emergency bus (UES) from switchgear 10. Start AOP-1 & JOP-1 as per requirement and monitor lube oil system parameters 11. Start AC SOP-1 and monitor seal oil system parameters
Methodology for re-establishing the DC Supply	Associate with EM- C&I for early restoration of 24 V DC Supply.

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Resp: SCE / AGM (EM) / AGM (C&I)

24V DC FAILURE IN CW PUMP HOUSE

DC Supply Configuration	24 V DC Supply for stage-2 CW pump house is having two independent 24 V DC boards provided with their own Battery Banks & Battery chargers. A Tie provision is available between both the boards in case of requirement. Both are not coupled.
Possible situations for Non-availability of DC Supply	Fire in Chargers/DCDB room
Annunciations / Alerts	All the signals at CWPH MMI will become bad
Consequences of DC Supply fail (Situation wise)	No command will get executed.
Steps for Safe Isolation / Shutdown of plant in case re-establishment of DC source is not feasible.	If the operator wants to close the valve and stop the CW motor, then it has to be done by pressing EPB & closing of BFV, As 24VDC is absent BFV solenoid will not be energized from local PB, BFV is to be closed by pressing close solenoid manually.
Methodology for re-establishing the DC Supply	Associate with EM for early restoration of 24 V DC Supply. Resp: SCE / DGM (EM) / DGM (C&I)
Recommendations / Modifications	NIL

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220 V DC Supply Failures in Switchyard and Substation Automation System

DC Supply Configuration	In this system two chargers are provided which are charging two independent battery banks and two sections of DCDB. The two sections of DCDB are connected with bus coupler. Normally the bus coupler remains in open condition.
Possible situations for Non-availability of DC Supply	Failure of both sections of DCDB
Annunciations / Alerts	1. 220 V DC Fail common alarm appears 2. Trip coil ½ unhealthy alarm appears
Consequences of DC Supply fail (Situation wise)	1. If both sections fails protections will not be available for outgoing lines breakers and GT/ST. 2. However, GCB protections are available from the Unit (for stage-1) and lines are protected from other ends. 3. No command will get executed from SAS system
Steps for safe Isolation / shutdown of plant in case re-establishment of DC source is not feasible.	Ensure DG sets of Units are kept in service 400KV Lines are to be tripped from other end. Resp: SCE / Switchyard Engr
Methodology for re-establishing the DC Supply	Associate with EM for early restoration of 220 V DC Supply. Resp: SCE / DGM (EM)
Recommendations / Modifications	NIL

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48V DC Failure for Power Line Carrier Communication (PLCC)

DC Supply Configuration	Two DC supply sources from independent Battery banks each with its Float cum Boost chargers. Source-1(265AH Battery) is having one charger and battery bank. Source-2 (265 AH) is having one charger and battery bank
Annunciations / Alerts	Main-1 Protection / Main-2 Protection Unhealthy alarm appears. Carrier Channel Lock out alarm failure appear in all Line Feeders Communication failure occurs
Possible situations for Non-availability of DC Supply	Failure of both DC Sources
Consequences of DC Supply fail (Situation wise)	Main 1& Main-2 scheme protection will not be in service. Communication with all substations will not be available. Inter trip & Direct trip for all protections from other substations will not occur. Data from Tejas network panel will stop reporting to SRLDC.
Steps for safe Isolation / shutdown of plant in case re-establishment of DC source is not feasible.	Establish communication with other Sub-stations for any emergency manual operation in case of requirement Resp: Shift Engr
Methodology for re-establishing the DC Supply	Inform EM and associate for restoration of DC at the earliest. Associate with EM for early restoration of 48 V DC Supply. Resp: SCE / AGM (EM)
Recommendations/ Modifications	NIL.

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DC supply for DG set for stage-1&2

DC Supply Configuration	DG-1, DG-2 and standby DG sets in both the stages. Each having one 24V Charger with two battery banks for cranking the machine. 24V DC supply for PLC
Possible situations for Non-availability of DC Supply	Fire in Chargers/24V DC PLC supply system
Annunciations / Alerts	DG set charger supply fail annunciation fail will come in DCS
Consequences of DC Supply fail (Situation wise)	No command will get executed and starting of DG set either auto or manual is not possible
Steps for Safe Isolation / Shutdown of plant in case re-establishment of DC source is not feasible.	Isolate the AC supply to the chargers
Methodology for re-establishing the DC Supply	Associate with EM for early restoration of DC Supply. Resp: SCE / DGM (EM) / DGM (C&I)
Recommendations / Modifications	NIL

220 Volts DC/110/24V Volts DC system of MUW area

DC Supply Configuration	24 V/110/220V DC Supply for MUW is having two independent DC boards provided with their own Battery Banks & Battery chargers. A Tie provision is available between both the boards in case of requirement. Both are not coupled.
Possible situations for Non-availability of DC Supply	Fire in Chargers/DCDB room Fault in chargers
Annunciations / Alerts	All the signals at MMI will become bad

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Consequences of DC Supply fail (Situation wise)	No command will get executed.
Steps for Safe Isolation / Shutdown of plant in case re-establishment of DC source is not feasible.	If the operator wants to stop the running drives then it has to be done by pressing the mechanical trip wherever breakers are used for Drives.
Methodology for re-establishing the DC Supply	Associate with EM & C&I for early restoration of 24/110/220 V DC Supply. Resp: SCE / DGM (EM) / DGM (C&I)
Recommendations / Modifications	NIL

220 Volts DC system of CHP area

DC Supply Configuration	- In Stage 1 HT switchgear and MCC 1 and 2 and in stage 2 HT switchgear and MCC 2A and 2B DCFB are supplied by this system. These DCFB are supplied from DCDB. DCDB having two incomers sourced with battery charger and battery bank. 220 Volts DCDB is in two sections, each section is provided with one battery charger and one set of battery bank. Normally the bus coupler remains in open condition. - For Stage 1 DCFB 3 and 4 at MCC 4 supply is coming from unit 1 220V DCDB # 1FA 03F- 03J and unit 2 220V DCDB # 2FA 10F- 10G - For stage 2 DCFB 5 and 6 at MCC 2C are supplied from unit 3 220V DCDB 6.17 R and unit 4 220V DCDB 9.11 R
Possible situations for DC Supply fail	Fire in Chargers/DCDB room Fault in chargers

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Consequences of DC fail	• All HT and LT breaker shall remain as it is. • LT conveyors motors can be operated from control room. • HT conveyors motor shall not be operated but scoop can be disengaged from control room.
Steps for Safe Isolation / Shutdown of plant in case re-establishment of DC source is not feasible.	• Necessary HT drives and incomers can be tripped from switchgear using manual push button. • Necessary LT incomers can be tripped from switchgear using manual push button.
Methodology for re-establishing the DC Supply	Restore DC supply at the earliest in association with EM Resp: SCE / DGM (EM)
Recommendations / Modifications	Nil

Failure of Un-interrupted Power Supply (UPS) in Stage 1 CHP

DC Supply Configuration	1) In this system there are two UPS chargers with one common battery bank of 400 volt. 2) The chargers are feeding supply to two inverters. Normally both the inverters are in service and if one inverter fails then other inverter takes the load. 3) If both the invertors fail, then the total UPS failure occurs. 4) For stage 1 CHP system two UPS are there. a) UPS in MCC1 caters load for main PLC, stage 1 control room RIO, RIO in MCC1, MCC 2, WTRIO (at MCC2) b) UPS in MCC4 caters load for RIO in MCC 3 and MCC4
Possible situations for Non-availability of UPS Supply	Failure of DC to both Inverters.

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Consequences of UPS fail	a.In case of failure of UPS in MCC1 both PLC will stop, and field equipment will lose communication with PLC. Drives cannot be operated from control room. b.In case of failure of UPS in MCC4 all drives from MCC 3 and 4 shall lose their communication with PLC and cannot be controlled from control room.
Steps for safe Isolation / shutdown of plant in case re-establishment of UPS source is not feasible.	a.To stop drives LT MCCs can directly tripped from hard button provided in control room. b.To stop HT drives local emergency stop to be used.
Methodology for re-establishing the UPS Supply	Restore UPS at the earliest in association with EM Resp: SCE / DGM (EM) / DGM (C&I)
Recommendations / Modifications	Nil

DC supply earth fault monitoring and rectification system.

Annunciations / Alerts	220V DC Earth fault annunciation will come in respective units DCS and alarm facia in Switchyard control room
<u>Monitoring</u>	Unit controllers in units and Offsite Engr in Switchyard control room
<u>Methodology</u>	Once earth fault reported will trace the source of fault and clear at the earliest. Resp: DGM (EM)
<u>Rectification system</u>	After identification same will be informed to controller of UCB. If possible identified module/feeder will be isolated for further checking taking the concerned area engineer into confidence.

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7. Implementation & Responsibility & Record keeping.

Activities to be carried out to reduce the failure in DC system.

Sl.No	Activity	Responsibility	Record Number
6.1	Implementation of LMI	Heads of EM, C&I and Operation	Not Applicable
6.2	Carrying out modifications	Heads of EM & C&I	Not Applicable
6.3	a. Preventive maintenance of 24V battery charger & 24V DCDB & 230V UPS system during O/H. OH frequency of unit linked is 2 years and offsite areas & MUW area & CHP area is 1 year.	Head of C&I	Document file number of C&I
6.4	a. Monthly PM checks on Batteries. b. Preventive maintenance of 220V & 110V battery charger & their DCDB during O/H. OH for Main plant is 2years and for MUW and Offsite area and CHP area is 1 year. c. Battery capacity test as per LMI /OD/OPS/ELEC/004	Head of EM	FC-1A2 FC-1A3 FC- 1A4,1A5,1A6, 1A7,1A8
6.5	a. Six monthly emergency exercises and training. b. Checking for any faults or annunciation at DCDB and	Head of Operation	Doc file numbers of Operation.

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	chargers of 220V DC & 48V of Main plant DC system shiftwise. c. Checking for any faults or annunciation at DCDB and chargers of 24V DC & 230V UPS system shiftwise. d. Checking for any faults or annunciation at distribution board cum chargers of 220V DC Offsite area (inside the plant) daily. e. Checking for any faults or annunciation at distribution board cum chargers of 220V DC & 110V DC & 24V DC at MUW area daily. f. Checking for any faults or annunciation at distribution board cum chargers of 220V DC & 110V DC & 24V DC at CHP area daily.		Operation Shift logbook Operation Shift logbook Daily operation logbook. Daily operation logbook. Daily operation logbook.
6.6	Overall Co-ordination	Head of EM	Not Applicable

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8. REVIEW

HOD(EMD) will review & take the approval of competent authority (HOP). The document will be reviewed on 2 yearly basis or earlier, if required.

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9. Checklist

Annexure-A

Steps for Safe isolation/shutdown of plant in case of re-establishment of Supply is not feasible

Unit 220 V DC Failure	
Steps for Safe Isolation / Shutdown of plant in case re-establishment of DC source is not feasible.	5. Make open announcement in the PA system and alert all concerned. Resp-Unit Control Engineer(UCE) 6. Manually trip GCB (for Unit-1 & 2 only) and field breaker from the local panel. Resp-UCE 7. Stop the necessary HT drives through switchgear manual push button. Resp- UCE 8. Identify the cause of DC failure and restore the same with the help of Elect Maint. Resp- UCE Ensure the following before opening 400KV main and tie breakers from 400KV Switchyard. 7. Start DG Set. Resp-TG Desk Engineer 8. Charge emergency switchgear through DG set by tripping the normal incomer breaker from IDA/2DA manually and closing the DG incomer manually from local breaker push buttons. Resp-TG Desk Engineer 9. Start Main Turbine AOP-I which is in Unit emergency bus. Resp-TG Desk Engineer 10. Start AC Seal Oil Pump-1 which is in Unit emergency bus. Resp-TG Desk Engineer 11. Start AOP- A of all TDBFPs which is in Unit emergency bus. Resp-TG Desk Engineer 12. Inform switchyard engineer to trip 400 KV side main and tie breakers. Resp- UCE and Switchyard Engr. This will result in total power failure in the unit with emergency switchgear in charged condition through DG set. Take actions as per the total power supply failure procedure with special care on emergency switchgear equipments as D.C. lube oil and seal oil pumps will not be available. Resp: UCE
Failure of Un-interrupted Power Supply (UPS) in Stage 1&2	
Steps for safe Isolation / shutdown of plant in case re-establishment of UPS source is not feasible.	3. Start main Turbine AOP/JOP / AC Seal Pumps from vertical control panel/UCD as direct start provision is available for them. Resp-TG Desk Engineer 4. Check the status of all critical equipments and ensure safe shutdown whenever required. Resp- UCE
Unit 24 V DC Failure Stage-I (2*500 MW Units)	
Steps for Safe Isolation / Shutdown of plant in case re-establishment of DC source is not feasible.	10. Main Turbine AOP-I, ACSOP-I & JOP-1 should be started from vertical panel as direct start provision is available for them. Resp-TG Desk Engineer

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	11. Monitor seal oil and lube oil systems parameters. Resp-TG Desk Engineer 12. Turbine is to be tripped from local trip gear in the governing rack. Resp-UCB 13. After tripping of turbine, Generator is to be tripped from GRP panel by activating GRP relay (low forward power relay 37 G1 or G2). Resp-TG Desk Engineer 14. PA fans are to be tripped on top most priority from local switchgear as remote tripping is not possible. Resp- Electrical operator 15. After ensuring the tripping of PA Fans, trip FD fans from local switchgear as remote tripping is not possible. Resp- Electrical operator 16. After tripping PA and FD fans, mills, CC Pumps, CEP, etc are to be tripped from local switchgear as remote tripping is not possible. Resp- Electrical operator 17. TDBFPs are to be tripped from local to avoid drum level high problem. Resp-TG Floor operator 18. TDBFP & Main Turbine barring gear valve to be opened from local. Resp-TG Floor operator
TG -24 V DC Failure Stage-II (2*500 MW Units)	
Steps for Safe Isolation / Shutdown of plant in case re-establishment of DC source is not feasible.	12. Start DC EOP from direct start push button given on control desk 13. Trip boiler by pressing Emergency trip buttons on Control Desk. All Mills, Feeders & both PA fans will trip. 14. Trip Turbine from local trip device in TG Governing Rack 15. Generator will trip on Low forward power protection after 30 seconds. Otherwise open both 400KV Breakers from ECD. 16. Stop BCW Pumps from MMI if drum level is too low. 17. TDBFPs are to be tripped from MMI to avoid drum level going high 18. Close MS Stop Valves MS-1&2 from MMI 19. Monitor condenser vacuum as both vacuum pumps inlet valves will close 20. Start DC JOP from local panel (Starter Cabinet) 21. Start AOP from Switch gear module Barring gear valves of main turbine & TDBFPs are to be opened from local
SG 24V DC Failure Stage-II (2*500 MW Units)	
Steps for Safe Isolation / Shutdown of plant in case re-establishment of DC source is not feasible.	8. All mills will trip on hard wired logic from UCD 9. Trip both PA Fans from MMI 10. Trip both FD fans from MMI to avoid furnace pressurization as both ID fans speed will be driven to minimum on 24V supply failure 11. Trip Turbine from Emergency Trip push button on control desk 12. Generator will trip on Low forward power protection 13. Monitor TG Lube Oil & Seal oil system parameters
STATION C&I 24 V DC Failure Stage-II (2*500 MW Unit)	

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Steps for Safe Isolation / Shutdown of plant in case re-establishment of DC source is not feasible.	12.ID fans VFDs speed will be driven to minimum and BFPs recirculation valves will open 13.Boiler will trip either on drum level low or furnace pressure high 14.Turbine will trip on MFT to turbine trip. 15.Generator will trip on low forward power protection 16. Both 11KV UA/UB buses will become dead resulting in loss of all unit auxiliaries. All Switchgear breakers will become inoperative from MMI. 17.Start DC EOP & DC SOP from direct start push buttons given on control desk. 18.Trip both TDBFPs from Local trip device. 19.Ensure ID/FD/PA fans & BCW Pumps are in tripped condition from switchgear. Otherwise trip them from switchgear emergency trip levers. 20.Start DG Set from Local and charge emergency bus (UES) from switchgear 21.Start AOP-1 & JOP-1 as per requirement and monitor lube oil system parameters Start AC SOP-1 and monitor seal oil system parameters
<u>24V DC FAILURE IN CW PUMP HOUSE</u>	
Steps for Safe Isolation / Shutdown of plant in case re-establishment of DC source is not feasible.	If the operator wants to close the valve and stop the CW motor, then it has to be done by pressing EPB & closing of BFV, As 24VDC is absent BFV solenoid will not be energized from local PB, BFV is to be closed by pressing close solenoid manually.
<u>220 V DC Supply Failures in Switchyard and Substation Automation System</u>	
Steps for safe Isolation / shutdown of plant in case re-establishment of DC source is not feasible.	Ensure DG sets of Units are kept in service 400KV Lines are to be tripped from other end.
<u>48V DC Failure for Power Line Carrier Communication (PLCC)</u>	
Steps for safe Isolation / shutdown of plant in case re-establishment of DC source is not feasible.	Establish communication with other Sub-stations for any emergency manual operation in case of requirement
<u>DC supply for DG set for stage-1&2</u>	
Steps for Safe Isolation / Shutdown of plant in case re-establishment of DC source is not feasible.	Isolate the AC supply to the chargers
<u>220 Volts DC/110/24V Volts DC system of MUW area</u>	
Steps for Safe Isolation / Shutdown of plant in case re-establishment of DC source is not feasible.	If the operator wants to stop the running drives then it has to be done by pressing the mechanical trip wherever breakers are used for Drives.
<u>220 Volts DC system of CHP area</u>	
Steps for Safe Isolation / Shutdown of plant in	Necessary HT drives and incomers can be tripped from switchgear using manual push button.

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case re-establishment of DC source is not feasible	Necessary LT incomers can be tripped from switchgear using manual push button.
Failure of Un-interrupted Power Supply (UPS) in Stage 1 CHP	
Steps for safe Isolation / shutdown of plant in case re-establishment of UPS source is not feasible.	a.To stop drives LT MCCs can directly tripped from hard button provided in control room. b.To stop HT drives local emergency stop to be used.